V.Imp # Hasse Diagram

Hasse Diagram is a visual representation / drawing used in mathematics to show the ordering of elements in a Partial Order Set (POSET) in the clean and simple way.

Steps To Design Hasse Diagram:

1. Start with a POSET - set of item and a valid ordering rule (like number divisibility or subset).

2. Remove self links & transitive links - Ignore the collection where items relate to itself and if $A \rightarrow B$ & $B \rightarrow C$ then remove the direct link between $A \rightarrow C$.

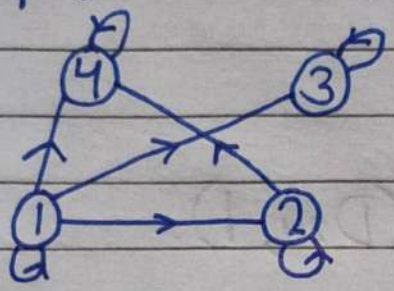
3. Draw & Arrange - Place smaller elements at the bottom and larger elements at the top, draw straight lines between elements where one immediately covers the other (elements / items are shown as a dot).

Here, arrowheads are not shown because direction is implicitly understood as moving upwards.

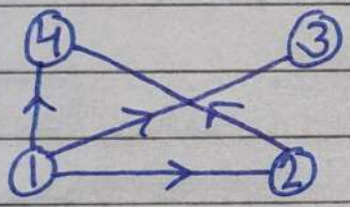
Q.1) Draw Hasse Diagram for $A=(\{1,2,3,4\}, |)$ with relation $R=\{(1,1),(1,2),(1,3),(1,4),(2,2),(2,4),(3,3),(4,4)\}$.

Soln:

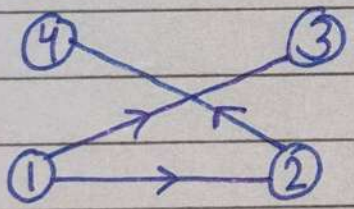
Create graph,



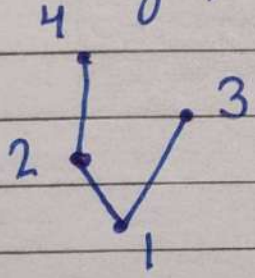
Remove self links,



Remove transitive links,



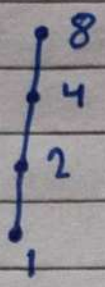
Draw & Arrange,



Q.2) Draw Hasse Diagram for $(D_8, |)$.

Sol^n:

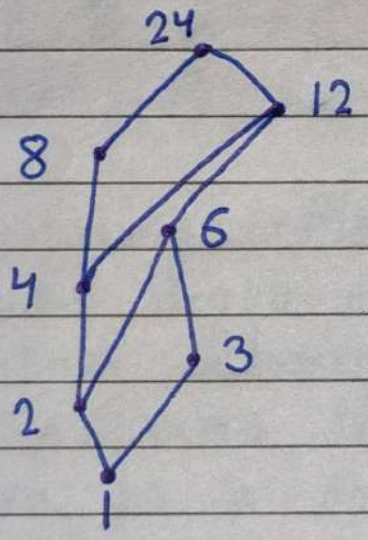
Set of divisors of 8 is $\{1, 2, 4, 8\}$ i.e. D_8.



Q.3) Draw diagram for $(D_{24}, |)$.

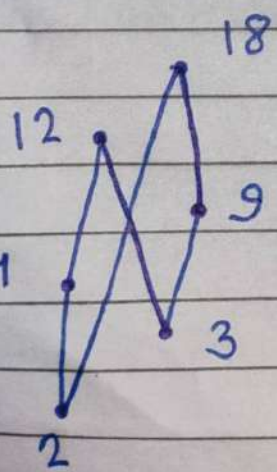
Sol^n:

$D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$



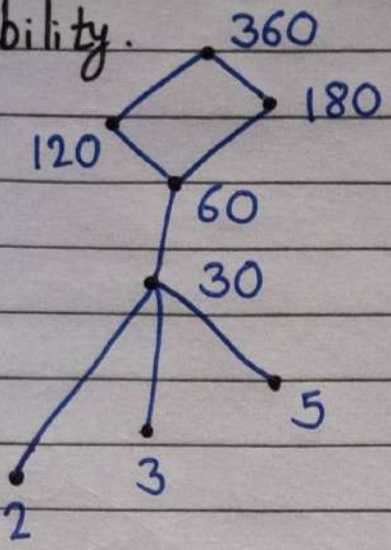
Q.4) Draw diagram for $A = \{2, 3, 4, 9, 12, 18\}$ under |.

Sol^n:



Q.5) Draw diagram for $A = \{2, 3, 5, 30, 60, 120, 180, 360\}$ under divisibility.

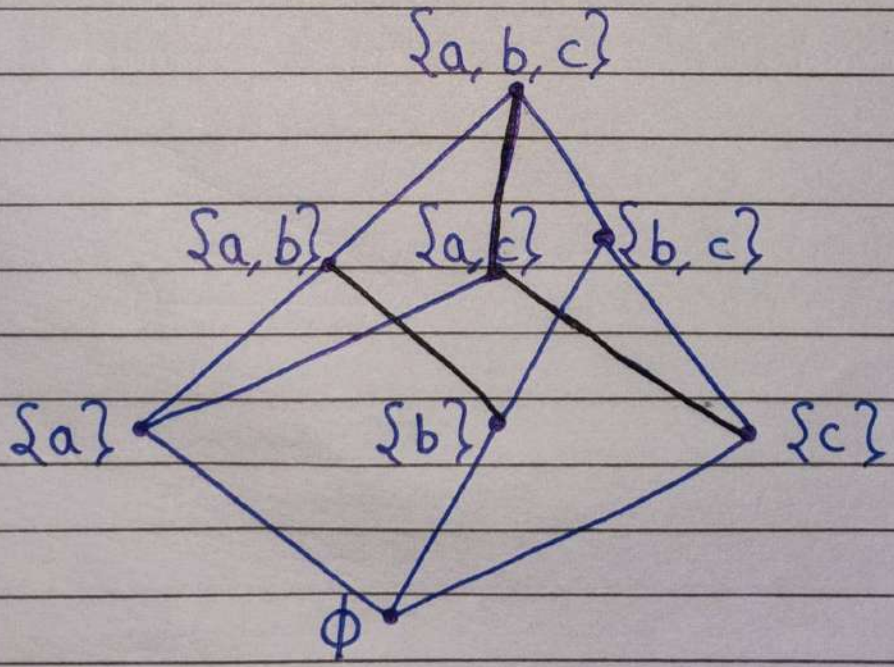
Sol^n:



Q.6) Draw hasse diagram for power set $(P(A), \subseteq)$ where $A = \{a, b, c\}$.

Sol^n:

$P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$



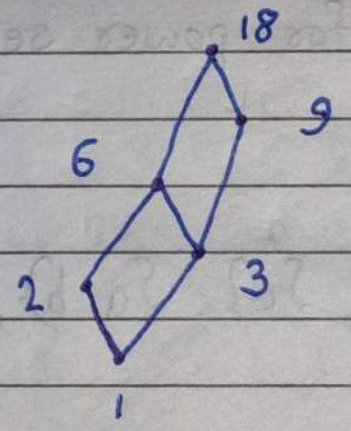
Lower Bound

An element $l \in A$ is called the lower bound of B if l is every bound of B i.e. $l \leq x$ for $x \in B$, where A & B are two sets $(B \subseteq A)$.

Example: Draw Hasse Diagram for D_{18} w.r.to divide operation.

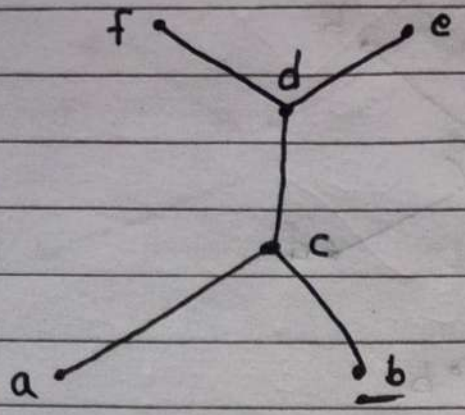
Soln:

$D_{18} = \{1, 2, 3, 6, 9, 18\}$



Lower Bound for $\{1, 2\} = \{1\}$
" " for $\{1, 3\} = \{1\}$
" " " $\{2, 3\} = \{1\}$
" " " $\{2, 6\} = \{1, 2\}$
" " " $\{2, 9\} = \{1\}$
" " " $\{2, 18\} = \{1, 2\}$
" " " $\{3, 6\} = \{1, 3\}$
" " " $\{3, 9\} = \{1, 3\}$
" " " $\{3, 18\} = \{1, 3\}$
" " " $\{6, 9\} = \{1, 3\}$
" " " $\{6, 18\} = \{1, 2, 3, 6\}$
" " " $\{9, 18\} = \{1, 3, 9\}$.

Q. Find lower bound for d & c, d & a, f & e, e & d a & b in the given Hasse Diagram?



Sol^n:

Lower bound for {d, c} = c, a & b.

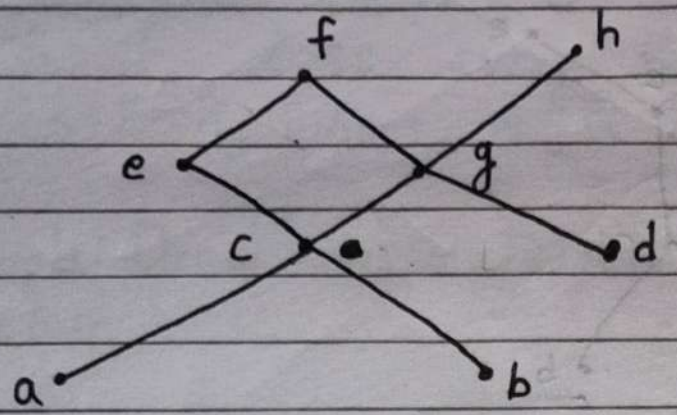
Lower bound for {d, a} = a.

Lower bound for {f, e} = d, c, a & b.

Lower bound for {e, d} = d, c, a & b.

Lower bound for {a, b} = {} or ϕ.

Q. Find the lower bounds for e & c, a & b, a & c, f & h, e & g, c & d in the given hasse diagram.



Soln:

L.B. for $\{e, c\} = \{c, a, b\}$

L.B. for $\{a, b\} = \{\bullet\}$ or ϕ

L.B. for $\{a, c\} = \{a\}$

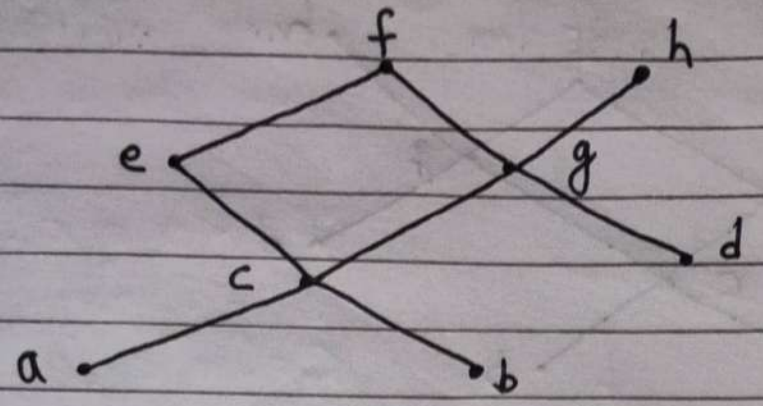
L.B. for $\{f, h\} = \{g, d, c, a, b\}$

L.B. for $\{e, g\} = \{c, a, b\}$

L.B. for $\{c, d\} = \{\}$ or ϕ.

Greatest Lower Bound (Meet or Infimum)

Q.



Soln:

G.L.B or Meet for $\{e, c\}$ = ~~{c}~~ c

G.L.B for $\{a, b\} = \phi$

G.L.B for $\{a, c\} = a$

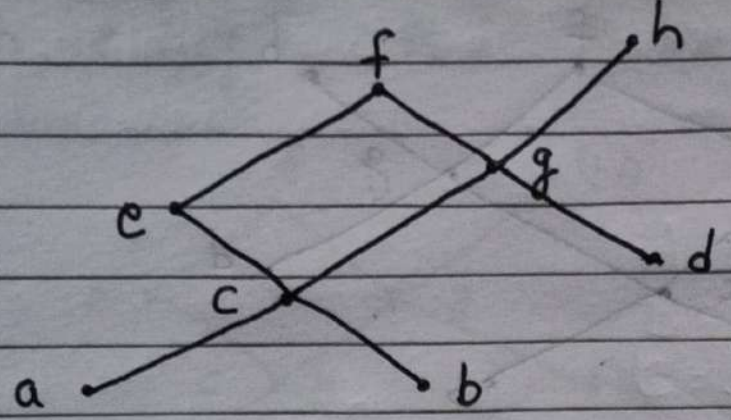
G.L.B for $\{f, h\} = g$

G.L.B for $\{e, g\} = c$

G.L.B for $\{c, d\} = \phi$.

Upper Bound

Q.



Soln:

U.B. for $\{e, c\} = \{e, f\}$

U.B. for $\{a, b\} = \{c, e, f, g, h\}$

U.B. for $\{a, c\} = \{c, e, g, f, h\}$

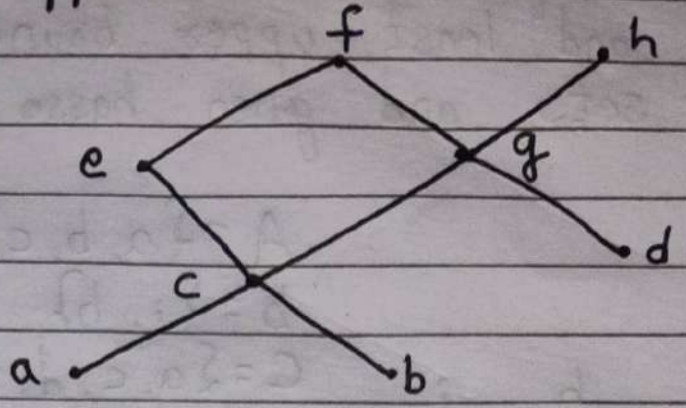
U.B. for $\{f, h\} = \phi$

U.B. for $\{e, g\} = \{f\}$

U.B. for $\{c, d\} = \{g, f, h\}$.

Least Upper Bound (Join) or (Supremum)

Q.



Soln:

L.U.B for $\{e, c\} = e$

L.U.B for $\{a, b\} = c$

L.U.B for $\{a, c\} = c$

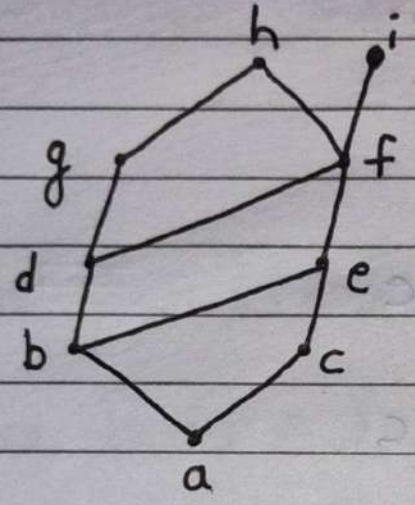
L.U.B for $\{f, h\} = \phi$

L.U.B for $\{e, g\} = f$

L.U.B for $\{c, d\} = g$.

Q. Find lower bound, greatest lower bound (meet), upper bound and least upper bound (join) for the following sets and given hasse diagram?

$A = \{a, b, c\}$
$B = \{i, h\}$
$C = \{a, c, d, f\}$



Soln:

For $A = \{a, b, c\}$,

L.B. $= \{a\}$
G.L.B. $= a$
U.B. $= \{e, f, h, i\}$
L.U.B. $= e$.

For $B = \{i, h\}$

L.B. $= \{f, d, e, b, c, a\}$
G.L.B. $= f$
U.B. $= \phi$
L.U.B. $= \phi$

For $C = \{a, c, d, f\}$

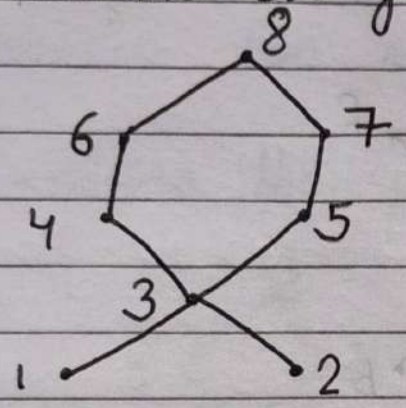
L.B. $= \{a\}$
G.L.B. $= a$
U.B. $= \{f, h, i\}$
L.U.B. $= f$.

Join Semi Lattice (∨)

In a POSET if least upper bound exist for every pair of element then the POSET is called join semi lattice. It is represented by '∨'.

Example: Determine the given is join semi lattice or not.

①



Soln:

L.U.B for {1, 2} = 3
L.U.B for {1, 3} = 3
~~L.U.B for {1, 4} =~~
L.U.B for {1, 4} = 4
L.U.B for {1, 5} = 5
L.U.B for {6, 8} = 8
⋮

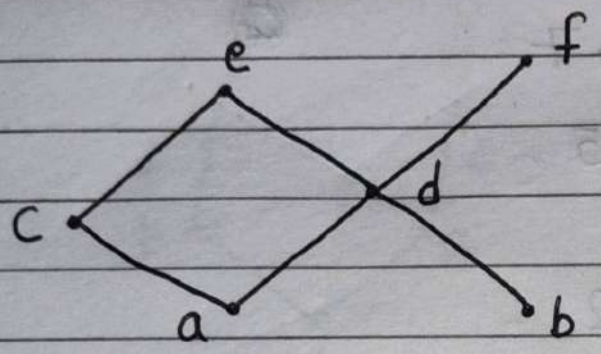
Since for every pair there is a least upper bound.

So, the given is join semi lattice.

~~Semi~~ Meet Semi Lattice ($\wedge$)

In a POSET if greatest lower bound exist for every pair of element then the POSET is called meet semi lattice.

Example: Determine the following is meet semi lattice or not.



Here, the G.L.B for $\{a, b\} = \phi$.

So, the given one is not meet semi lattice.

Imp # Lattice

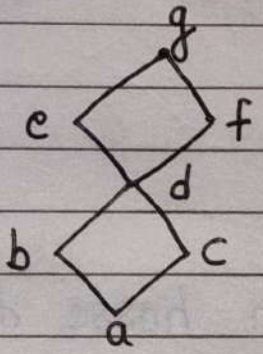
A POSET is called lattice if every pair of elements has a least upper bound (join) and a greatest lower bound (meet).

Lattice is a mathematical structure with two binary operation Join($\vee$) and Meet($\wedge$).

Types: Bounded, Complemented, Modular, Complete, Distributed, Boolean

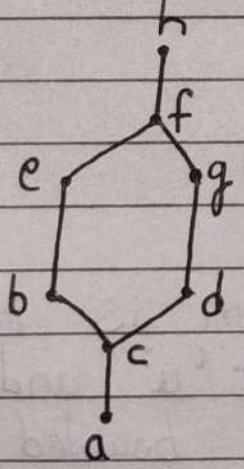
Example: Determine the given is lattice or not.

①



Here, for all pairs there is least upper bound and greatest lower bound. So, the given is lattice.

②



Here, for all pairs there is join and meet.
So, it is lattice.

Properties of Lattice

Idempotent Laws

$$a \underset{\text{(Meet)}}{\wedge} a = a$$

& $$a \underset{\text{(Join)}}{\vee} a = a$$

Commutative Laws

$$a \wedge b = b \wedge a$$

$$a \vee b = b \vee a$$

Associative Laws

$$a \wedge (b \wedge c) = (a \wedge b) \wedge c$$

$$a \vee (b \vee c) = (a \vee b) \vee c$$

Absorption Laws

$$a \wedge (a \vee b) = a$$

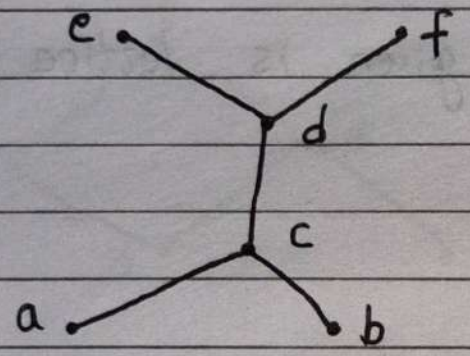
$$a \vee (a \wedge b) = a$$

Bounded Lattice $[0 \leq x \leq 1 \quad \forall x \in \text{POSET}]$

(least element ↓, greatest element ↑)

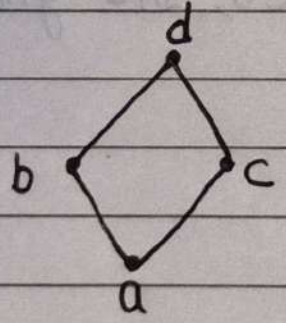
A POSET $(S, \wedge)$ is called a bounded lattice iff every finite set of element has join and meet, and if it contains a least element (denoted as 0) such that $0 \leq x \ \forall \ x \in S$ and a greatest element (1) such that $x \leq 1 \ \forall \ x \in S$.

Example: Determine the given is bounded lattice or not.

① 

Here, ~~{e, f}~~ the given hasse diagram is not lattice. So, not bounded lattice.

②



The given hasse diagram is lattice and also has a least element 'a' and a greatest element 'd'. So, it is bounded lattice.

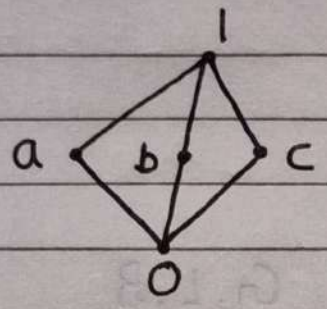
Modular Lattice

A lattice is known as modular lattice if

$$a \vee (b \wedge c) = (a \vee b) \wedge c$$

(∨ : Join, ∧ : Meet)

Example: Determine the given hasse diagram is modular lattice or not.



$b \wedge c = 0$ $\qquad$ $a \vee b = 1$

$a \vee 0 = 1 \wedge c$

$\Rightarrow a \neq c$

Thus, the given is not modular lattice but if $a \leq c$ then the given is modular lattice.

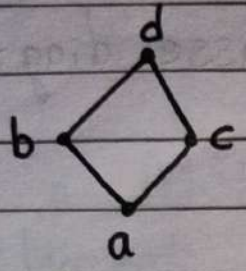
Complete Lattice

A POSET is a complete lattice if every subset of it has both least upper bound (join) and greatest lower bound (meet).

Complemented Lattice

A lattice $(L, \leq)$ is a complemented lattice if it is a bounded lattice and every element has at least one complement.

Q.



Maximum = d

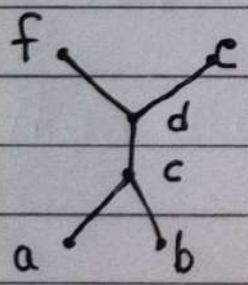
Minimum = a

	L.U.B	G.L.B
{a, b}	b	a
{b, c}	d	a
{b, d}	d	b
{c, d}	d	c
{a, c}	c	a
{a, d}	d	a

Thus, the given is complemented lattice.

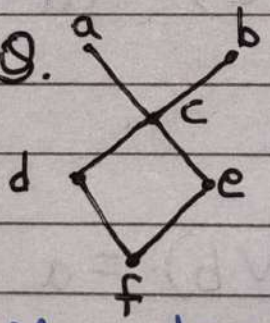
Maximum Element: If only one maximal exist then it is maximum (for finite sets).

Minimum Element: If only one minimal exist then it is minimum (for finite sets).

Q. 

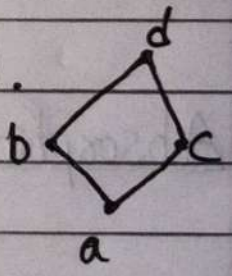
Maximal = f, e
Minimal = a, b

Maximum = ϕ
Minimum = ϕ

Q. 

Maximal = a, b
Minimal = f

Maximum = ϕ
Minimum = f

Q. 

Maximum = d
Minimum = a